Answer Key - Computer Science - Graduate Level Assessment 12 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** C
   Options: A) They are generated when memory cycles are "stolen".; B) They are used in place of data channels.; C) They can indicate completion of an I/O operation.; D) They cannot be generated by arithmetic operations.
   Note: Interrupts serve as signals to the processor indicating that an I/O operation has completed, allowing the CPU to efficiently manage tasks and respond promptly to events without constant polling.

2. **Answer:** B
   Options: A) Installing and configuring a IDS that can read the IP header; B) Comparing the TTL values of the actual and spoofed addresses; C) Implementing a firewall to the network; D) Identify all TCP sessions that are initiated but does not complete successfully
   Note: Comparing the Time to Live (TTL) values of the actual and spoofed addresses helps detect IP address spoofing because legitimate packets typically have a consistent TTL value that reflects their time in transit, while spoofed packets may exhibit irregular TTL values indicating they did not originate from the expected source.

3. **Answer:** B
   Options: A) No, I cannot compute the key.; B) Yes, the key is k = m xor c.; C) I can only compute half the bits of the key.; D) Yes, the key is k = m xor m.
   Note: The correct answer is based on the principle of the One-Time Pad (OTP) encryption, which states that the ciphertext (c) is generated by performing a bitwise XOR operation between the plaintext message (m) and the key (k), allowing the key to be easily derived as k = m xor c.

4. **Answer:** C
   Options: A) Replaced; B) Overview; C) Changed; D) Violated
   Note: A hash function provides a unique fixed-size output for a given input, allowing any alteration of the message to result in a different hash value, thereby ensuring the integrity of the message by detecting changes.

5. **Answer:** C
   Options: A) Active, inactive, standby; B) Open, half-open, closed; C) Open, filtered, unfiltered; D) Active, closed, unused
   Note: The correct answer is "Open, filtered, unfiltered" because these terms specifically describe the states of ports as identified by Nmap during a network scan, indicating whether a port is accessible, blocked by a firewall, or not definitively categorized.

True / False

6. **Answer:** True
   Options: A) True; B) False
   Note: Snort is considered true because it is an open-source network intrusion detection system that effectively monitors network traffic in real-time, allowing for the detection of malicious activities and policy violations.

7. **Answer:** False
   Options: A) True; B) False
   Note: CFB, or Cipher Feedback mode, is not a block cipher itself but rather an operating mode for block ciphers, which means it relies on an underlying block cipher for encryption rather than being a standalone encryption algorithm.

8. **Answer:** False
   Options: A) True; B) False
   Note: Snort is primarily an intrusion detection and prevention system (IDS/IPS) rather than a comprehensive network analysis tool, which limits its effectiveness in multiprotocol diverse networks.

9. **Answer:** False
   Options: A) True; B) False
   Note: ARPANET was an early packet-switching network developed in the late 1960 s primarily for research and communication among academic and government institutions, whereas Freenet is a modern decentralized network focused on anonymous file sharing and privacy.

10. **Answer:** False
    Options: A) True; B) False
    Note: The statement is false because in a complete K-ary tree of depth N, the ratio of nonterminal nodes to total nodes is actually (K^N - 1)/(K^N - 1 + 1), which does not simplify to (K-1)/K.

Long Form Response

11. **Answer:** def find_literals(text, pattern): | return (match.re.pattern, s, e)
    Note: correct because it outlines a function that utilizes regular expressions to search for a specified pattern within a given string and returns the relevant match details, such as the start and end positions of the found pattern.

12. **Answer:** def tn_gp(a,n,r): | return tn
    Note: The function correctly implements the formula for the t-nth term of a geometric series, given the first term (a), the term number (n), and the common ratio (r), which is essential for calculating any term in the series.

End of Answer Key